

AWS EventBridge Scheduler Hands On Lab

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon EventBridge Scheduler** was used to stop an **Amazon EC2** instance automatically at a fixed interval. First, a custom **IAM role** was created because EventBridge Scheduler needs permission to call EC2 actions, and the normal AWS service-role list did not include the scheduler trust relationship directly. A custom trust policy for scheduler.amazonaws.com was used, and **AmazonEC2FullAccess** was attached for the demo. Then a schedule named SC test was created using a **rate expression** of every 2 minutes. The target action was set to **StopInstances** for the selected EC2 instance named **CRM system**. After the schedule ran, the instance changed from running to stopped, proving that EventBridge Scheduler can automate timed AWS actions.

Understanding EventBridge Scheduler

Amazon EventBridge Scheduler lets you run AWS actions automatically at a future time or on a repeating interval. It exists so you can automate routine tasks like stopping a server, starting a server, sending a message, or triggering a workflow without manual intervention. In this lab, the scheduler was connected to EC2 so it could stop a specific instance on a repeating timer. Because the scheduler needs permission to act on AWS resources, an IAM role with a custom trust policy is required.

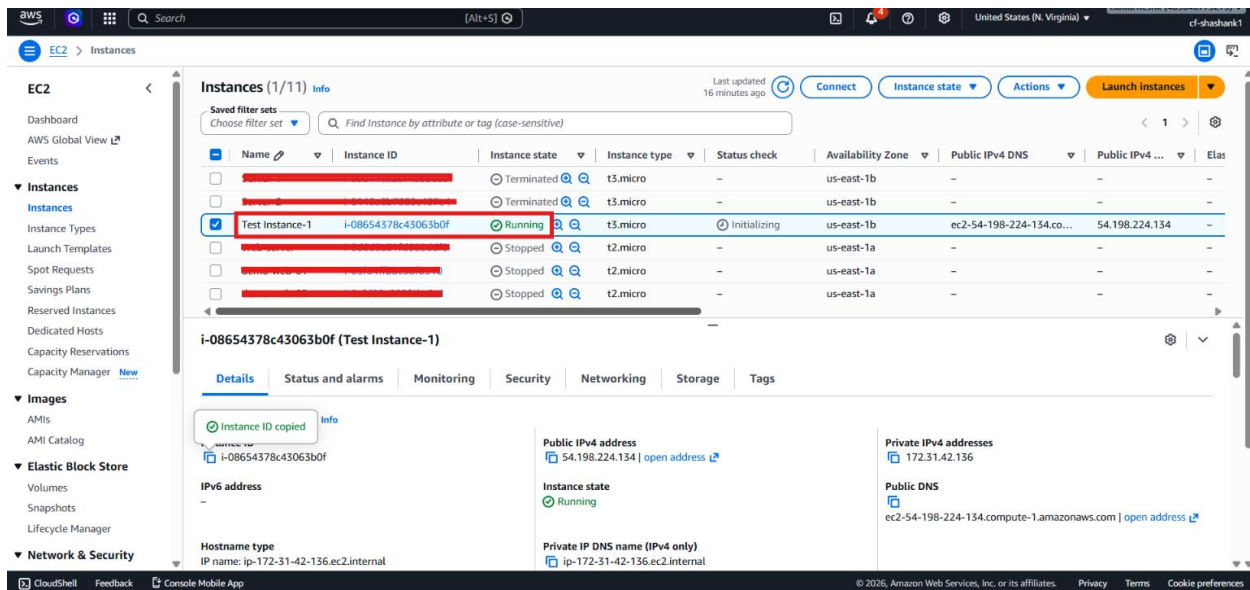
Example:

A team has a development EC2 instance that should run only during working hours. Instead of stopping it manually every day, they create an EventBridge schedule that runs every 2 minutes for testing, then later adjust it to a real business schedule. The schedule uses an IAM role to stop the instance automatically.

Lab Steps

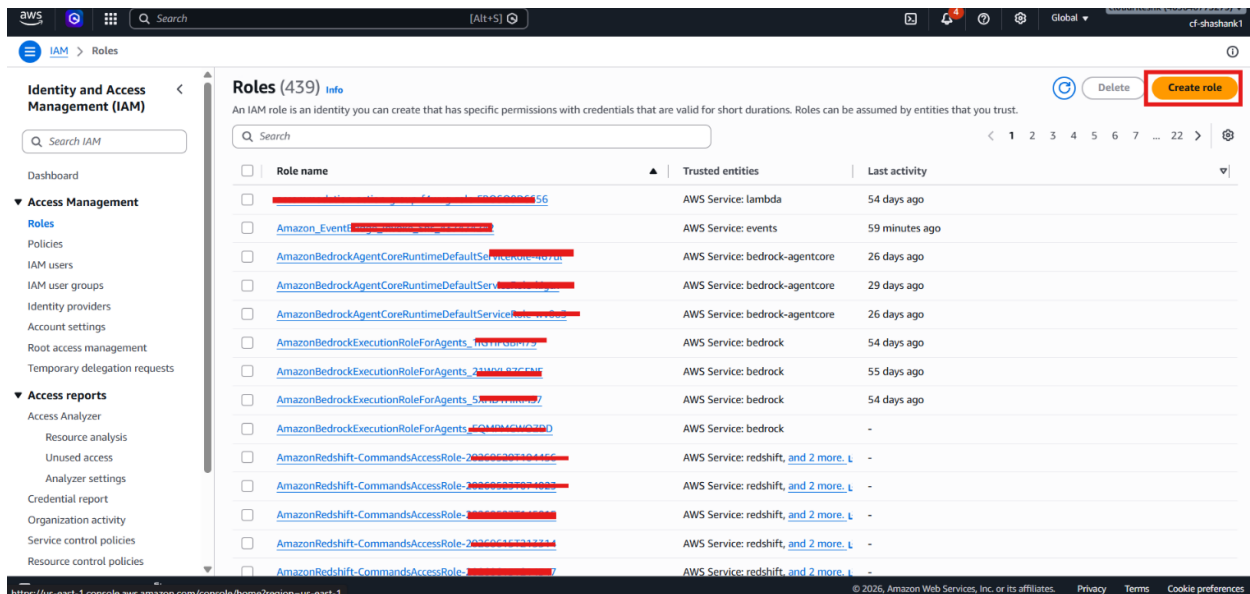
Step 1 – Review the EC2 Instance

- Sign in to the AWS Management Console.
- Open **Amazon EC2**.
- Confirm that the target instance is in the **running** state.



Step 2 – Create the IAM Role for Scheduler

- Open IAM.
- Go to Roles.
- Click Create role.



- Do not choose a normal AWS service role for this case.
- Use a **custom trust policy**.
- Add the trust relationship for: scheduler.amazonaws.com
- Continue to permissions.

Trusted entity type

- ☐ AWS service
- ☐ AWS account
- ☐ Web identity
- ☐ SAML 2.0 federation
- ☒ Custom trust policy

Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```

1 *
2 {
3   "Version": "2012-10-17",
4   "Statement": [
5     {
6       "Effect": "Allow",
7       "Principal": {
8         "Service": "scheduler.amazonaws.com"
9       },
10      "Action": "sts:AssumeRole"
11    }
12  ]
  
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

- Attach the required policy for the demo: AmazonEC2FullAccess

Add permissions

Permissions policies (1/2321)

Choose one or more policies to attach to your new role.

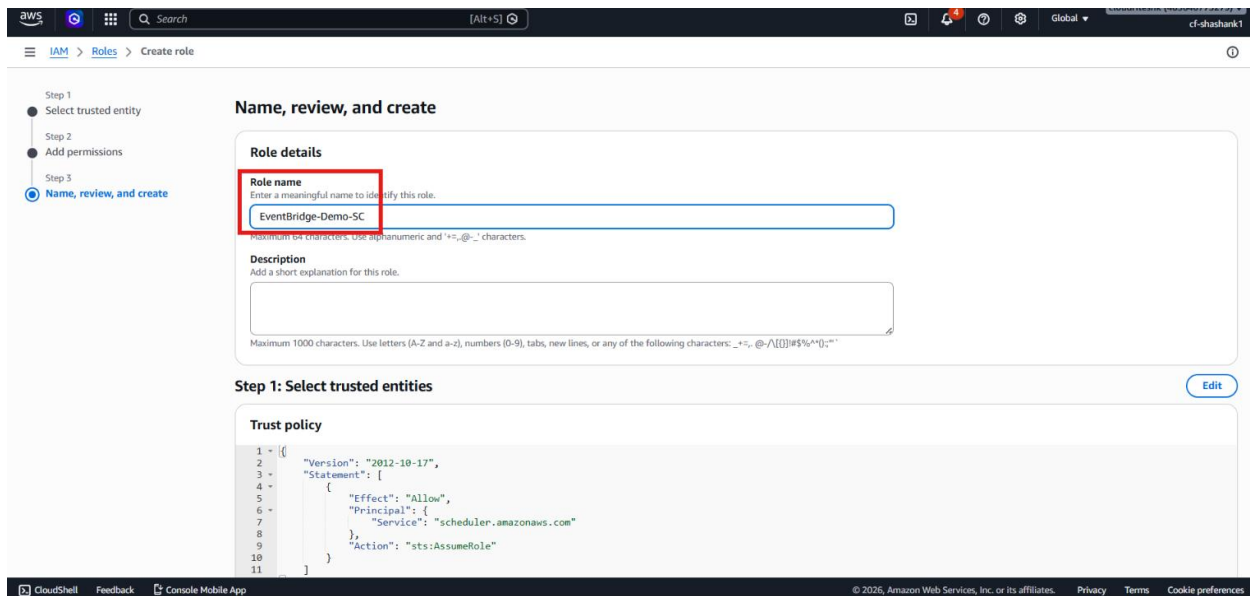
Filter by Type: All types (1 match)

Policy name	Type	Description
☒ AmazonEC2FullAccess	AWS managed	Provides full access to Amazon EC2 via th...

Set permissions boundary - optional

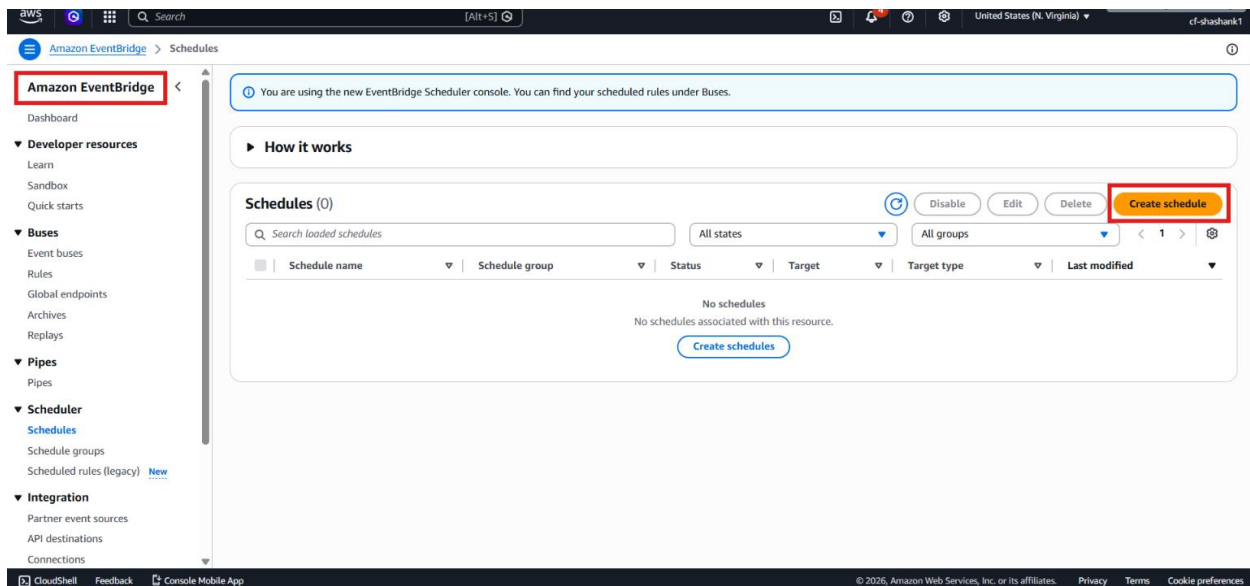
[Cancel](#) [Previous](#) [Next](#)

- Enter the role name: EventBridge-Demo-SC
- Create the role.



Step 3 – Open EventBridge Scheduler

- Open Amazon EventBridge.
- Go to Scheduler.
- Click Create schedule.



Step 4 – Configure the Schedule

- Enter the schedule name: SC-Test
- Keep the default schedule group.
- Choose **Recurring schedule**.

Specify schedule detail

Schedule name and description

Schedule name
SC-Test
Use only letters, numbers, dashes, dots or underscores. Max 64 characters.

Description - optional
Enter description
Maximum of 512 characters.

Schedule group
Each schedule needs to be placed in a schedule group. By default, a schedule is placed in the 'Default' group. You can also [create your own schedule group](#). You can only add tags to a schedule group, not a schedule.
default

Schedule pattern

Occurrence [Info](#)
You can define an one-time or recurrent schedule.

☐ One-time schedule ☒ Recurring schedule

Time zone
The time zone for the schedule.
(UTC+05:30) Asia/Calcutta

- Choose a **Rate-based** schedule.
- Set the interval to: every 2 minutes
- Leave flexible window disabled for this lab.

☐ One-time schedule ☒ Recurring schedule

Time zone
The time zone for the schedule.
(UTC+05:30) Asia/Calcutta

Schedule type
Choose the schedule type that best meets your needs.

☐ Cron-based schedule
A schedule set using a cron expression that runs at a specific time, such as 8:00 a.m. PST on the first Monday of every month.

☒ Rate-based schedule
A schedule that runs at a regular rate, such as every 10 minutes.

Rate expression [Info](#)
Enter a value and the unit of time to run the schedule.
rate (2 minutes)
Value Unit

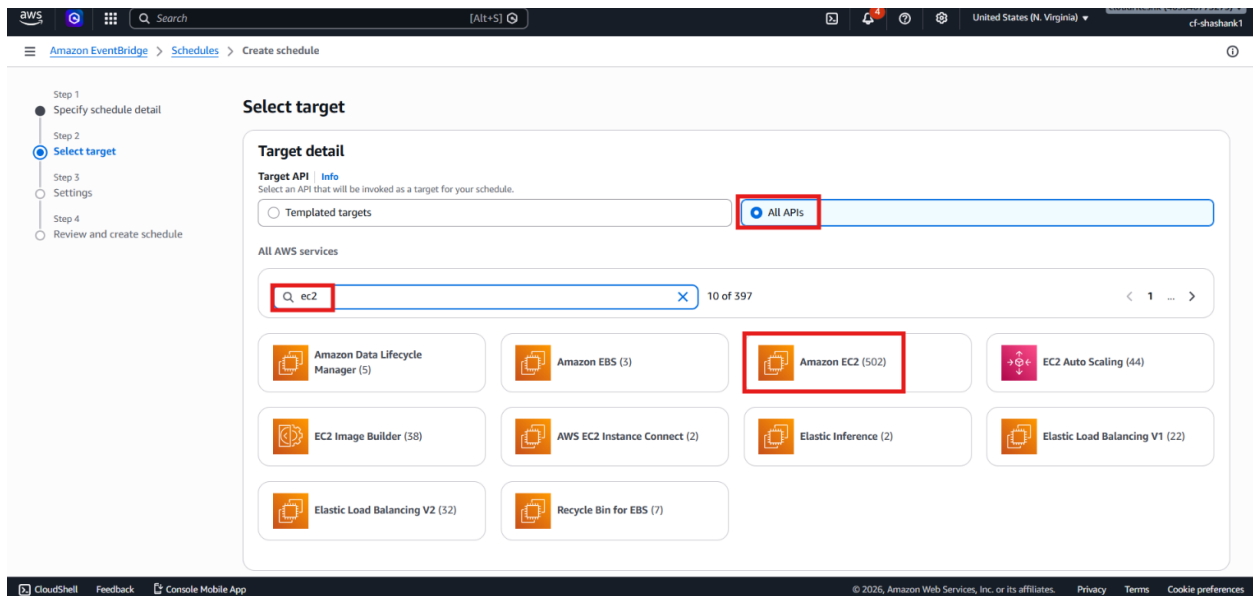
Flexible time window
If you choose a flexible time window, Scheduler invokes your schedule within the time window you specify. For example, if you choose 15 minutes, your schedule runs within 15 minutes after the schedule start time.
Off

Timeframe

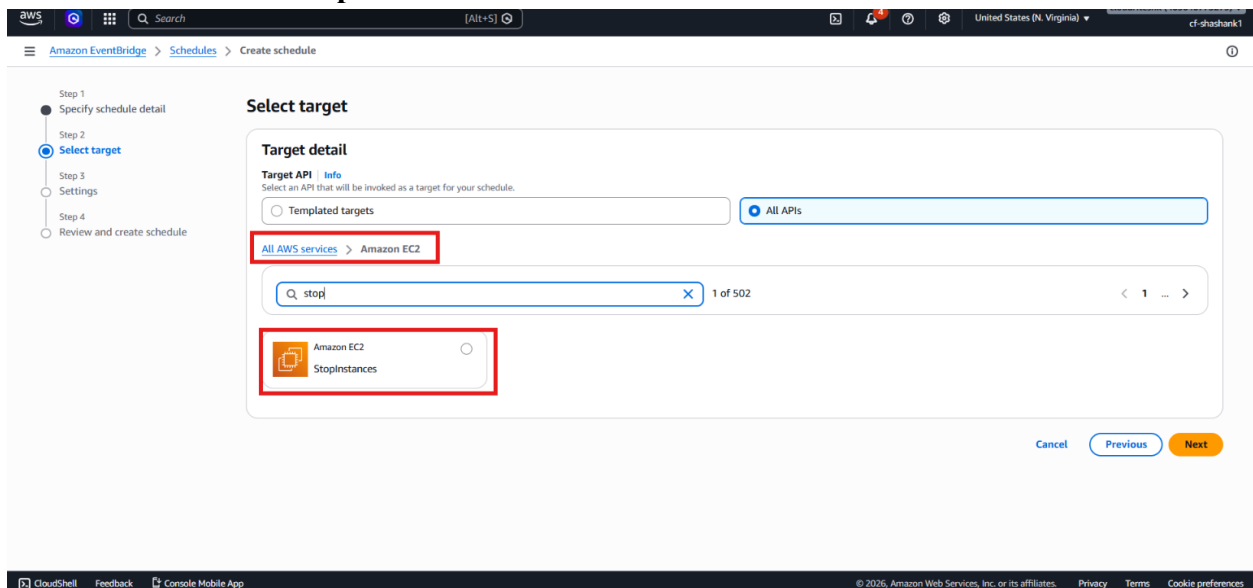
Daylight saving time
Amazon EventBridge Scheduler automatically adjusts your schedule for daylight saving time. When time shifts forward in the Spring, if a cron expression falls on a non-existent date, your schedule invocation is skipped. When time shifts backwards in the Fall, your schedule runs only once and does not repeat its invocation. The following invocations occur normally at the specified date and time.

Step 5 – Select the Target Action

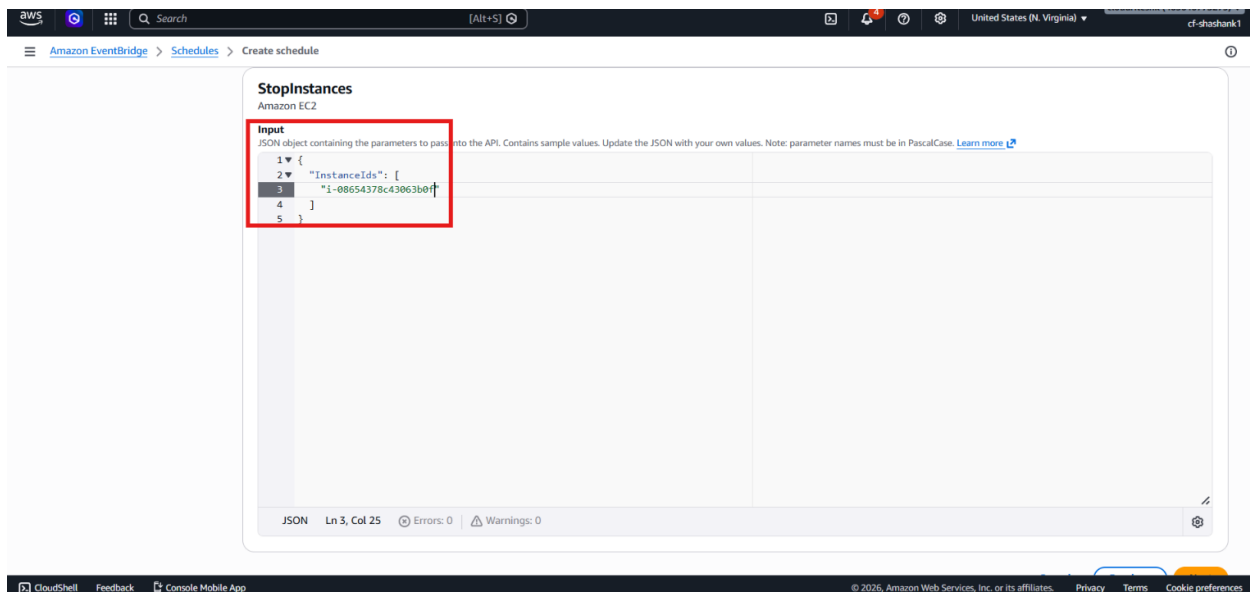
- In the target section, choose **All APIs** and search the EC2 target option.
- Select **EC2**.



- Choose the action **StopInstances**.



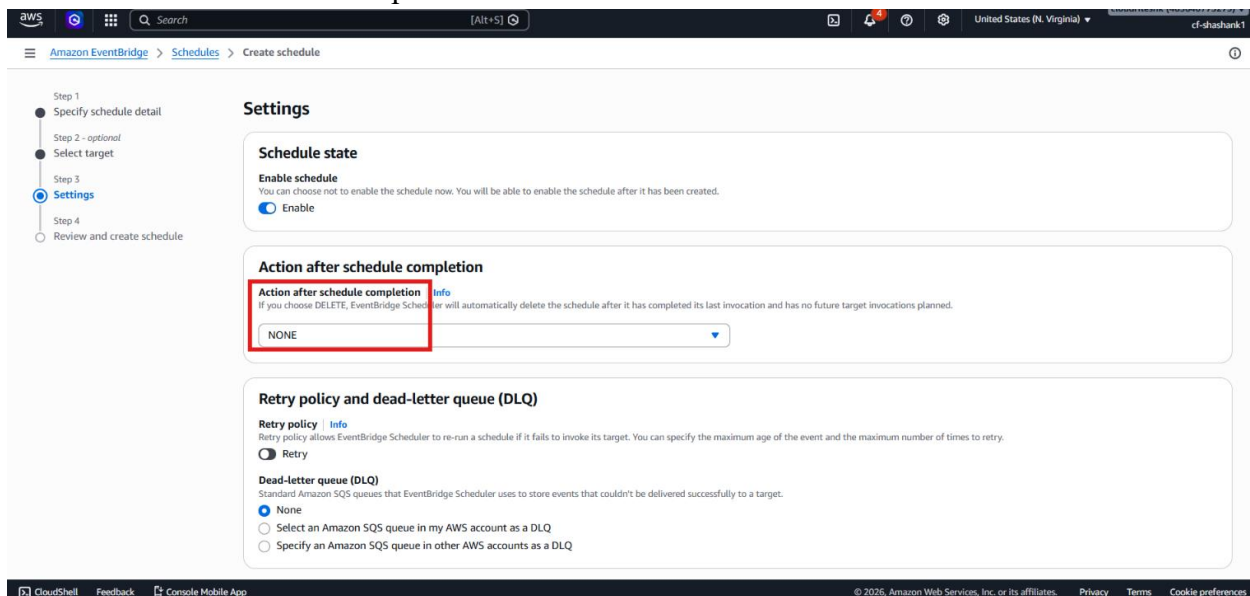
- Enter the EC2 instance ID for **Test Instance-1**.



- Confirm the selected instance is the correct target.

Step 6 – Configure Permissions

- Action after schedule completion: NONE.



- Select the IAM role created earlier.

Create schedule

Standard Amazon SQS queues that EventBridge Scheduler uses to store events that couldn't be delivered successfully to a target.

☒ None
☐ Select an Amazon SQS queue in my AWS account as a DLQ
☐ Specify an Amazon SQS queue in other AWS accounts as a DLQ

Encryption info
 By default, EventBridge Scheduler encrypts event metadata and message data that it stores under an AWS owned key (encryption at rest). EventBridge Scheduler also encrypts data that passes between EventBridge Scheduler and other services using Transport layer Security (TLS) (encryption in transit).
 Your data is encrypted by default with a key that AWS owns and manages for you. To choose a different key, customize your encryption settings.
☐ Customize encryption settings (advanced)

Permissions info
 EventBridge Scheduler requires permission to send events to the target, and based on the preferences you select, integrate with other AWS services such as AWS KMS and Amazon SQS.

Execution role
☐ Create new role for this schedule
☒ Use existing role

Select an existing role
 EventBridge-Demo-SC

Cancel Previous **Next**

- Confirm that the scheduler will use that role to stop the EC2 instance.
- Leave DLQ and extra options unchanged unless required.

Step 7 – Create the Schedule

- Review all settings.
- Click **Create schedule**.

Create schedule

Schedule state and permissions

Schedule state	Enabled	Execution role	EventBridge-Demo-SC
Action after schedule completion	NONE		

Retry policy and dead-letter queue (DLQ)

Retry policy	Max age of event: -	Retry policy	Maximum retries: -
Dead-letter queue ARN	None		

Encryption

Customer master key (CMK)	aws/scheduler	Description	Default master key that protects my Amazon EventBridge Scheduler data when no other key is defined
Key ARN	-		

Cancel Previous **Create schedule**

- Wait for the schedule to run at the configured interval.
- Return to **Amazon EC2** and confirm that the instance changes to **stopped**.

The screenshot displays the AWS Management Console for the EC2 service. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows a list of instances. The first instance, 'Test Instance-1' with ID 'i-08654378c43063b0f', is highlighted with a red border. Its state is 'Stopped', which is also highlighted with a red box. Below the table, the details for 'Test Instance-1' are shown, confirming its state as 'Stopped'.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elast
Test Instance-1	i-08654378c43063b0f	Stopped	t3.micro	-	us-east-1b	-	-	-
...	...	Stopped	t2.micro	-	us-east-1a	-	-	-
...	...	Stopped	t2.micro	-	us-east-1a	-	-	-
...	...	Stopped	t2.micro	-	us-east-1a	-	-	-
...	...	Stopped	t2.micro	-	us-east-1a	-	-	-

i-08654378c43063b0f (Test Instance-1)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary

Instance ID: i-08654378c43063b0f

IPv6 address: -

Hostname type: IP name: ip-172-31-42-136.ec2.internal

Public IPv4 address: -

Instance state: Stopped

Private IP DNS name (IPv4 only): ip-172-31-42-136.ec2.internal

Private IPv4 addresses: 172.31.42.136

Public DNS: -

Verification

- The IAM role with a custom trust policy was created successfully.
- EventBridge Scheduler was configured with a rate-based schedule.
- The schedule targeted the EC2 **StopInstances** action.
- The **CRM system** instance stopped automatically after the schedule ran.
- The lab confirmed that EventBridge Scheduler can automate EC2 operations.